

Discovery Executive Summary

Project: discovery-14-aug · Generated: 14/08/2026, 11:54:42

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Architecture & Design Analysis	—

1. Architecture & Design Analysis

EXECUTIVE SUMMARY

The Klearcom platform is a monorepo with three layers: a Laravel backend (997 LOC across 6 controllers, 4 models, 2 services), a React 19 / TypeScript SPA frontend (983 LOC across 4 pages, 4 components, 1 hook, 1 store), and a Node.js/Express dev-api (832 LOC). The architecture exhibits several moderate-to-high-risk hotspots. The most severe issue is the complete absence of a Repository layer — all 25+ Eloquent ORM access points are scattered directly through controllers and services with no data-access abstraction. Additional concerns include business logic embedded in controllers (reachability-percentage calculations duplicated across `ConnectController`, `LegacyReportController`, and `RealTimeTestService`), cross-domain model access in `LegacyReportController` and `DashboardController`, and on the frontend, oversized page components with mixed concerns plus a legacy class component with an intentional resource leak. The dominant risk is change amplification: modifying the reachability formula requires synchronized edits across 3 backend files and 1 dev-api file. **Layers covered:** Backend (6 PHP controllers, 4 models, 2 services, 1 legacy mapper — 18 PHP source files), Frontend (4 pages, 4 components, 1 hook, 1 store, 1 API client, 1 types file — 12 TS/TSX/JSX source files), Dev-API (5 JS source files).

1.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	74 avg (6 controllers); but <code>LegacyReportController</code> contains duplicated business logic, <code>extract()</code> , and KPI math	Moderate
H2	Missing Service Layer	Controllers accessing repos/models directly	<10	10–20	>20	14 direct model access points across controllers	Moderate
H3	Missing Repository Pattern	Direct DB/ORM access points outside repositories	<10	10–20	>20	25+ (0 repository classes; all ORM access is direct)	High Risk
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	0	Good
H5	Shared Utility Abuse	Utility files w/ business logic	0	1–5	>5	1 (<code>LegacyDataMapper</code> with unsafe <code>extract()</code>)	Moderate
H6	Direct SQL in Controllers	ORM compliance %	>90%	60–90%	<60%	100% (no raw SQL; all queries use Eloquent)	Good
H7	God Classes	Classes >1000 LOC	0	1–3	>3	0	Good
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	4 (LegacyReportController imports Connect + Discovery models; DashboardController imports both domains)	Moderate
H9	Shared Database Coupling	Tables shared across domains	<10%	10–30%	>30%	0% (4 tables with clear domain ownership)	Good
F1	Business Logic in Components	Avg LOC per component	<150	150–300	>300	113 avg across 8 components/pages; but <code>ConnectPage</code> =222 LOC, <code>DiscoveryPage</code> =176 LOC	Moderate

F2	Missing Frontend Service/Data Layer	Components w/ inline API calls	<10	10–20	>20	2	Good
F3	God / Oversized Components	Components >400 LOC	0	1–3	>3	0	Good
F4	Prop Drilling / Global State Abuse	Max prop-drilling depth	≤2	3–4	>4	1 level	Good
F5	Legacy / Inconsistent Component Patterns	Legacy-pattern components	0	1–10	>10	1 (<code>LegacyMonitorPoller.jsx</code> — class component with interval leak)	Moderate

1.4 Actions Required

Hotspot	Action	Rating	Priority
H3 — Missing Repository Pattern	Create <code>ConnectMonitorRepository</code> , <code>ConnectCheckResultRepository</code> , <code>DiscoveryJobRepository</code> , <code>DiscoveryNodeRepository</code> with interfaces; bind in <code>AppServiceProvider</code> ; migrate all 25+ direct Eloquent calls to use repositories	High Risk	Critical
H1 — Fat Controllers	Extract reachability calculation into <code>ReachabilityCalculator</code> service; remove duplicate <code>buildTree()</code> from <code>LegacyReportController</code> ; replace <code>extract()</code> with explicit array access	Moderate	High
H2 — Missing Service Layer	Create <code>ConnectService</code> , <code>DiscoveryService</code> , <code>DashboardService</code> ; move CRUD and KPI logic from controllers into services; replace hardcoded KPIs in <code>DashboardController</code>	Moderate	High
F5 — Legacy / Inconsistent Patterns	Convert <code>LegacyMonitorPoller</code> class component to function component with <code>useEffect</code> cleanup; add Error Boundary around <code>LegacyDashboardWidget</code> ; rename <code>.jsx</code> to <code>.tsx</code>	Moderate	High
H5 — Shared Utility Abuse	Replace <code>extract()</code> calls in <code>LegacyDataMapper</code> with explicit destructuring; register class in container for DI instead of <code>new</code> instantiation	Moderate	Medium
H8 — Domain Boundary Violations	Move models into domain namespaces (<code>App\\Modules\\Connect\\Models</code> , <code>App\\Modules\\Discovery\\Models</code>); create domain service interfaces for cross-domain queries	Moderate	Medium
F1 — Business Logic in Components	Extract React Query hooks into per-domain custom hooks; centralize query invalidation in <code>useRealtimeTest</code> ; split <code>ConnectPage</code> and <code>DiscoveryPage</code> into sub-components	Moderate	Medium

1.5 Expected Outcomes

- **Separation of concerns:** Controllers become thin HTTP translators; all business logic lives in testable service and domain classes, eliminating the 4-location reachability formula duplication.
- **Testability:** Repository interfaces enable unit testing with in-memory fakes; services can be tested without HTTP or database overhead.
- **Independent module evolution:** Bounded contexts with published interfaces allow Connect and Discovery teams to evolve their schemas and logic independently, paving the way for microservice extraction.
- **Frontend maintainability:** Custom hooks and sub-components keep page files under 100 LOC; centralized invalidation removes duplicated cache management across pages.
- **Reduced regression risk:** Eliminating `extract()` , fixing the interval leak, and adding Error Boundaries remove three categories of runtime bugs (variable injection, memory leaks, unhandled crashes).

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